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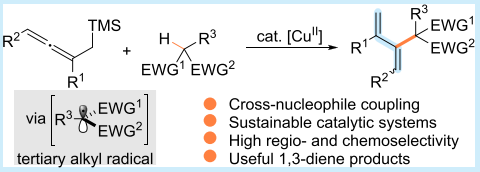
Copper-Catalyzed Cross-Nucleophile Coupling of β-Allenyl Silanes with Tertiary C−H Bonds: A Radical Approach to Branched 1,3-Dienes

Qi-Chao Shan, Lu-Min Hu, Wei Qin, and Xu-Hong Hu*

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ABSTRACT: Described herein is a distinctive approach to branched 1,3-dienes through oxidative coupling of two nucleophilic substrates, β-allenyl silanes, and hydrocarbons appending latent functionality by copper catalysis. Notably, C(sp^3)−H dienylation proceeded in a regiospecific manner, even in the presence of competitive C−H bonds that are capable of occurring hydrogen atom transfer process, such as those located at benzylic and other tertiary sites, or adjacent to an oxygen atom. Control experiments support the intermediacy of functionalized alkyl radicals.

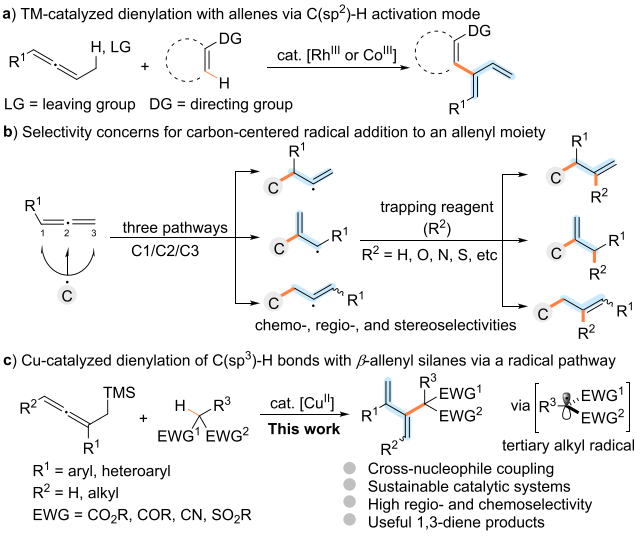


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The ongoing pursuit of efficient and selective methods to access synthetically useful reagents or intermediates by chemists has greatly influenced the design, synthesis, and applications of complex molecular scaffolds in chemical sciences. Allenes are a versatile class of unsaturated substrates in transition-metal-catalyzed coupling reactions due to their high and unique reactivity.[1] The presence of two orthogonal cumulative π bonds in allenes endows them with diverse characteristics to be qualified as precursors of alkenyl, allenyl, allyl, and 1,3-dienyl moieties, as well as in cyclization reactions.[2] Among them, dienylation represents an attractive strategy for the efficient incorporation of valuable 1,3-diene functionality.[3] Earlier studies in this field typically relied on organometallic reagents coupled with allenes,[4] largely impeding their economy and applicability. Later, elegant breakthroughs via metal-catalyzed aryl or alkenyl C−H dienylation with allenes have been independently realized by Glorius, Fu, Kapur and Volla,[5] where preinstalled directing groups were required to control the regioselectivity (Scheme 1a). Despite these achievements, oxidative couplings of C(sp^3)−H bonds[6] with allenes toward dienylation still remain a formidable challenge in terms of selectivity and reactivity. In this regard, we seek out an appealing radical approach to access alkyl functionalized 1,3-dienes from readily available allene and alkane substrates by considering the rich precedence of oxidative radical transformations.[7]

The addition of a carbon-centered radical to π-systems of allenes is arguably among the most powerful and straightforward methods to forge C−C bonds for increasing the molecular complexity.[8] It has been established that reactive vinyl or allylic radical species can be generated through a selective radical attack to an allene at three positions. The stability of the in situ formed radicals was crucial to the following radical-trapping steps, which mostly ended up with

Scheme 1. Carbofunctionalization of Allenes via C−H Activation and Radical Strategies



the formation of substituted alkenes via hydro-[9] and difunctionalization,[10] as well as annulation processes (Scheme 1b).[11] In those transformations manifold, the issue of chemo-, regio-, and stereoselectivities has been a persistent concern. We envisioned that the relatively more stable allyl radical species

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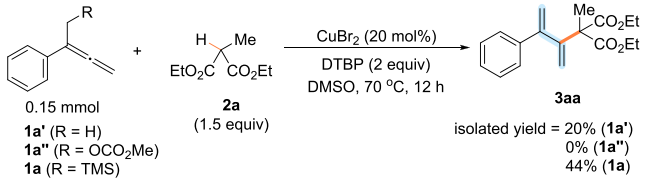
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could undergo oxidation and subsequent β-elimination to build conjugated dienes. To our best knowledge, a method targeting privileged 1,3-dienes through selective addition of carbon-centered radicals to allenes has never been disclosed, despite their versatile applications in organic synthesis.[12]

1,3-Dicarbonyl compounds have been recognized by many research groups including us as the sources of functionalized sp^3 carbon-centered radicals via single electron oxidation,[13] and recently, our laboratory has documented their use in the tertiary alkylation of alkenes and alkynes.[14] Features of note in our protocols include the following: (a) efficient generation of tertiary carbon-centered radicals from aliphatic C−H bonds,[15] instead of prefunctionalized substrates, such as halides, alcohols, carboxylic acids, etc.;[16] (b) incorporation of functionalized alkyl moieties holding the options for derivatizations; (c) sustainable catalytic systems (using Cu or Fe) without the need for expensive ligand. Because of the ubiquitous nature and high importance of conjugated dienes, we wonder whether this principle could be engaged in radical-mediated functionalization of reactive allenes to produce 1,3-dienes. Described herein are the discovery, exploration, and application of an unprecedented copper-catalyzed oxidative coupling of allenes with tertiary C−H bonds (Scheme 1c). The protocol gives rise to the regio- and chemoselective formation of highly functionalized 1,3-dienes, thus leaving ample room for further synthetic manipulations.

To assess the feasibility of our expected transformation, we synthesized a set of allene substrates affiliated with different α-alkyl substituents, and tested their reactivity (Scheme 2). By

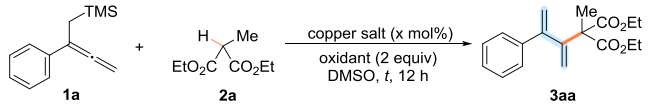
Scheme 2. Reaction Discovery



employing commercial diethyl methylmalonate (**2a**) as an alkyl radical precursor, the coupling product **3aa** was produced in 20% yield from α-methyl allene (**1a′**) isomerization[17] by using $CuBr_2$ as the catalyst and di-*tert*-butyl peroxide (DTBP) as the oxidant. Replacement of the methyl group at the α position of allene with carbinol carbonate (**1a″**) led to no conversion. The negative outcome implied that nucleophilic allenylation of malonate might not be involved in this reaction.[18] To our delight, exposure of allenylmethylsilane (**1a**) to the identical conditions gave a 44% yield of **3aa**. It is known that β-allenyl silanes often act as versatile nucleophiles which tend to attack carbonyls, iminiums, or other reactive electrophilic agents via an S_E2' pathway.[19] This newly developed methodology unlocks the potential utility of such silicon-based carbon π-systems for the installation of 1,3-diene motifs in cross-nucleophile coupling reactions.[20]

Once determining the β-allenyl silane **1a** as the model substrate, we centered on evaluating the reaction parameters with low-cost copper salts. Through extensive screening of copper salts with different counterions and oxidant states (Table 1, entries 1−6), it was found that $CuBr_2$ showed the best performance. Extra addition of phosphine (entry 7) or nitrogen (entry 8) ligands did not improve the conversion. The reaction appeared to be sensitive to the temperature

Table 1. Evaluation of Reaction Conditions for Cu-Catalyzed Tertiary Alkylation of β-Allenyl Silane[a]



entry	copper salt	x	t (°C)	yield (%)[b]
1	$CuBr_2$	20	70	44
2	$CuCl_2$	20	70	18
3	$Cu(OAc)_2$	20	70	trace
4	CuBr	20	70	28
5	CuI	20	70	42
6	CuCN	20	70	trace
7[c]	$CuBr_2$	20	70	38
8[d]	$CuBr_2$	20	70	37
9	$CuBr_2$	20	80	19
10	$CuBr_2$	20	60	47
11	$CuBr_2$	20	50	13
12	$CuBr_2$	10	60	39
13	$CuBr_2$	40	60	54
14	$CuBr_2$	50	60	52
15[e]	$CuBr_2$	40	60	31
16[f]	$CuBr_2$	40	60	68
17[f,g]	$CuBr_2$	40	60	70

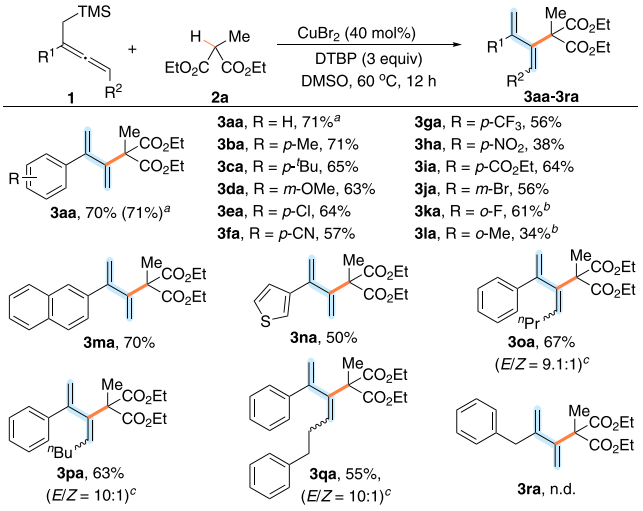
[a]Unless otherwise noted, all reactions were conducted with **1a** (0.15 mmol), **2a** (0.225 mmol), copper salt (20 mol %), and oxidant (0.3 mmol) in DMSO (1.5 mL) under N_2 for 12 h. [b]Isolated yield. [c]PCy_3 (20 mol %) was added. [d]2,2-Bipyridyl (20 mol %) was added. [e]Cy_2NH (0.3 mmol) was added. [f]The reaction was performed using **1a** (0.21 mmol) and **2a** (0.15 mmol). [g]DTBP (3 equiv) was used.

conducted. A dramatically reduced yield of **3aa** was observed at the elevated temperature (80 °C), which was ascribed to the substantial decomposition of **1a** (entry 9). The minimum appropriate temperature was ascertained to be 60 °C, at which point **3aa** was obtained in an undiminished yield (entries 10 and 11). Increasing the amount of $CuBr_2$ resulted in an enhancement to the reactivity (entries 12−14). The addition of Cy_2NH as base, which usually played a beneficial role in the β-elimination process, proved to be deleterious to the reaction at this time (entry 15). Employing **1a** in an excess of stoichiometry (1.4 equiv) with **2a** as a limiting reagent afforded **3aa** in 68% yield (entry 16), which was further improved to 70% upon an increase of DTBP loading (entry 17). An additional survey of oxidant and solvent choices indicated that DTBP and DMSO were uniquely effective (see Tables S1 and S2 for details).

The established optimal conditions stimulated us to explore the scope and limitation of various β-allenyl silanes with **2a** as the coupling partner. A satisfactory yield of **3aa** (71%) was maintained while enlarging the reaction to a 1 mmol scale. As shown in Scheme 3, a wide range of allenes bearing diverse substituents on the phenyl ring could be converted to functionalized 1,3-dienes in moderate to good yields under simple copper catalysis (**3ba**−**3la**). Synthetically useful functional groups, such as halides (**3ea**, **3ja**, and **3ka**), nitrile (**3fa**), trifluoromethyl (**3ga**), nitro (**3ha**), and ester (**3ia**), were well tolerated. Substitution patterns with electronic differentiation had an inapparent effect on the efficiency in most cases. However, decorating sterically encumbered substituents (F and Me) on the *ortho* position of the phenyl ring caused a drop in reaction rate and by means of raising the temperature, the corresponding products **3ka** and **3la** could be furnished in 61%

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Scheme 3. Variation of β-Allenyl Silanes

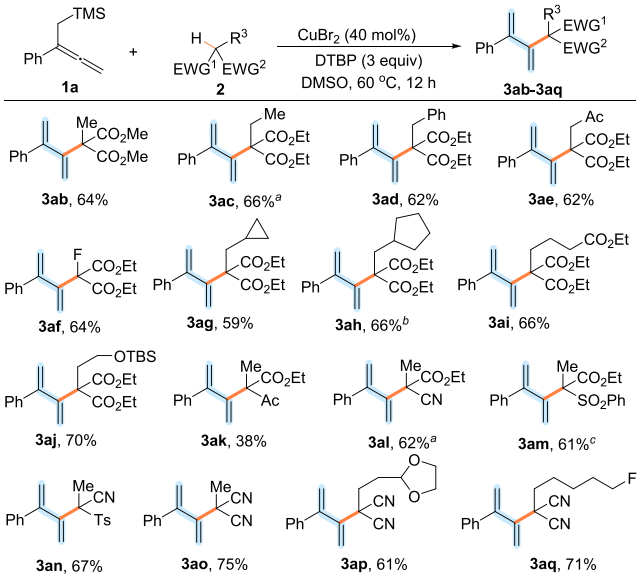


[a]On 1 mmol scale. [b]At 80 °C. [c]The E/Z ratio of stereoisomers was determined by ^{1}H NMR analysis of the crude product. n.d. = not detected.

and 34% yields, respectively. Extended π-systems, such as a naphthyl-conjugated allene, proved to be the viable substrate, smoothly delivering the desired product **3ma** in 70% yield. Aside from aryl-substituted allenes, a heteroaromatic substrate also followed the transformation, as evidenced by the facile production of thiophene-tethered 1,3-diene **3na** in 50% yield. Several allenylmethylsilanes flanked by alkyl chains were well accommodated, providing the access to the trisubstituted 1,3-dienes (**3oa**–**3qa**) in 55–67% yields with high regio- and stereoselectivities. Unfortunately, benzyl-substituted allene, as the example of less activated aliphatic allenes, was incompatible with this coupling under the present conditions (**3ra**).

Next, we investigated the chemical diversity of tertiary C(sp^3)–H bonds that could be selectively functionalized in Cu-catalyzed dienylation reactions (Scheme 4). The use of

Scheme 4. Variation of Tertiary C(sp^3)–H Bonds

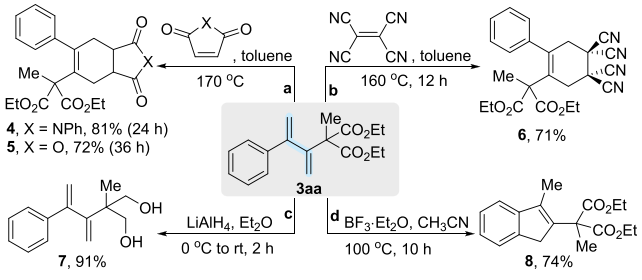


[a]At 70 °C. [b]At 80 °C. [c]At 50 °C. Ts = tosyl.

dimethyl methylmalonate did not alter the reactivity, affording **3ab** in 64% yield. Pleasingly, the coupling partners could be successfully extended to α-substituted malonates appended with ethyl, benzyl, keto, fluoro, cycloalkyl, ester, and silyl ether groups, yielding the desired products (**3ac**–**3aj**) with similar efficiency. It should be mentioned that substrates possessing competitive aliphatic C–H bonds that are able to undergo a hydrogen atom transfer process, for example of those located at benzylic and other tertiary sites, α-keto carbon center, or even adjacent to an oxygen atom, are all applicable in these oxidative coupling reactions. Remarkably, the scope of C(sp^3)–H bonds could be extended to analogous methine carbons attached to other electron-withdrawing groups, such as ketone, nitrile, and arylsulfonyl functionalities. The coupling allowed the assembly of tertiary alkyl substituted 1,3-dienes in up to 75% yield (**3ak**–**3ao**). In addition, the parent 2-alkyl malononitriles reacted with **1a** in a chemospecific manner (**3ap** and **3aq**), further underlining the robustness of this protocol.

We then sought to illustrate the postsynthetic utility of the resultant 1,3-dienes through a series of derivatization reactions (Scheme 5). Intermolecular Diels–Alder reactions of buta-

Scheme 5. Derivatization Reactions



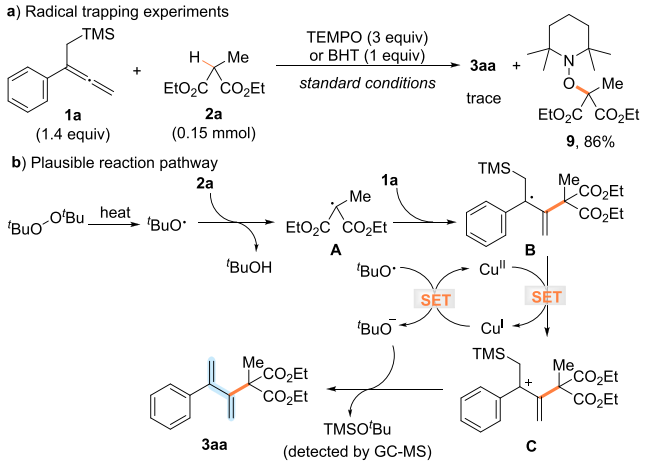
diene **3aa** with several dienophiles, such as *N*-phenylmaleimide, maleic anhydride, and tetracyanoethylene in toluene, provided cycloadducts **4**–**6** in excellent yields. Moreover, the 1,3-diester moiety of **3aa** could be readily transformed into 1,3-diol **7** upon $LiAlH_4$ reduction. The value of aryl substituted 1,3-diene was further showcased by the downstream preparation of indene **8** via a Lewis acid mediated intramolecular cyclization.[21]

Finally, several control experiments were carried out to elucidate the mechanism. No alkylated product was detected in the absence of either copper salt or DTBP oxidant. Subsequently, radical trapping experiments were performed with 3 equiv of 2,2,6,6-tetramethylpiperidine *N*-oxide (TEMPO) or 1 equiv of 2,6-di-*tert*-butyl-4-methylphenol (BHT) being added to the standard conditions. As expected, the coupling between **1a** and **2a** was strongly suppressed. Instead, the adduct **9** of alkyl radical trapping was formed in 86% yield, suggesting the involvement of alkyl radical intermediates in this event (Scheme 6a).

On the basis of the above results and literature precedence,[13,14,22] we propose a plausible reaction pathway shown in Scheme 6b. The coupling starts by thermal homolysis of DTBP to generate *tert*-butoxyl radicals, which abstracts a hydrogen atom from **2a** to form the electrophilic 1,3-diester radical species **A**. When the previous observations in catalysts screening are considered, it can be rationalized that coordination of $CuBr_2$ as a Lewis acid with the diester functionality may occur to facilitate the generation of species **A**.[23] Subsequently, radical addition to the central sp-hybridized

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Scheme 6. Control Experiments and Proposed Mechanism



carbon of **1a** forms an allylic radical intermediate **B**, which is simultaneously stabilized by benzylic conjugation. The presence of an aryl group had a crucial effect on the stability of the radical species according to the inactivity of allene **2r**. The radical **B** is oxidized by Cu^{II} to produce an allylic carbocation **C** along with Cu^{I} via a single electron transfer (SET) process. Meanwhile, a second SET event from *tert*-butoxyl radical occurs to regenerate Cu^{II} species. Intermediate **C** undergoes a β-Si elimination with the aid of a *tert*-butoxyl anion to deliver the final product **3aa**. Clearly, formation of a side-product TMSO^{t}Bu of the process can be validated by GC-MS analysis (see the Supporting Information).

In conclusion, a mechanistically distinct, practical strategy for the cross-nucleophile coupling of β-allenyl silanes with tertiary C−H bonds has been described. By using simple copper catalytic systems without the use of a ligand or additive, the reaction provides a regio- and chemoselective approach to value-added branched 1,3-dienes harboring a diverse collection of functionalized tertiary alkyl motifs, which were usually tortuous to access. Mechanistic investigations suggest the intermediacy of tertiary alkyl radicals. Product modification allowing for the assembly of carbocycle scaffolds further broadened the chemical space of this methodology. We anticipate the potential utility of silicon-based unsaturated molecules being reported in the context of radical chemistry.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02112.

Experimental procedures, characterization data, and NMR spectra for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

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